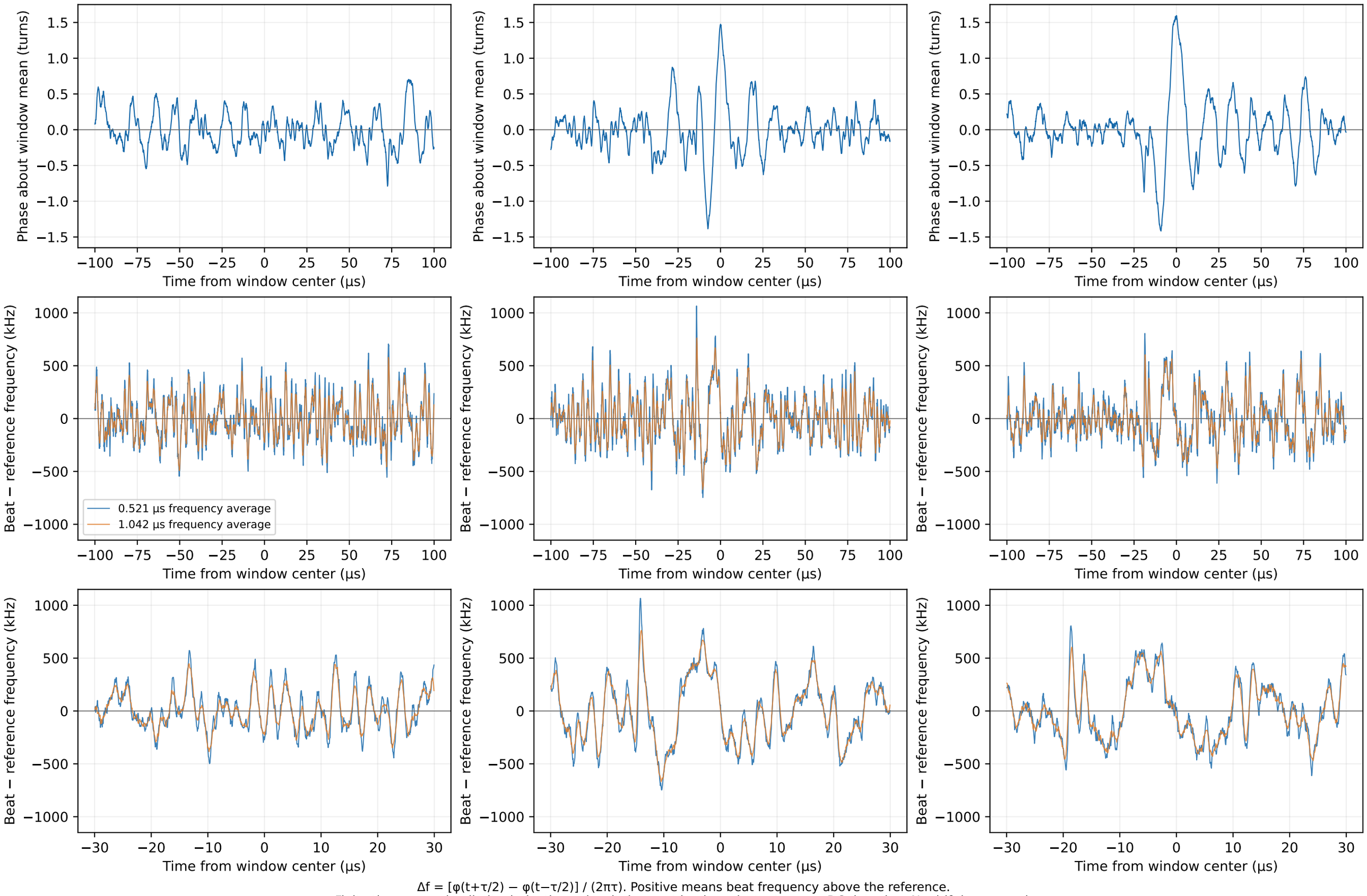
Capture 3: unwrapped phase converted to instantaneous frequency error

Ordinary interval
t = 1.050000 ms

First range-crossing event
t = 2.501172 ms

Largest phase excursion
t = 10.742448 ms



Δf = [φ(t+τ/2) − φ(t−τ/2)] / (2πτ). Positive means beat frequency above the reference.
Finite time averaging limits derivative noise; it does not imply an instantaneous DC detuning. No drift is removed.